Eternal Product Analyst OA — IIT Dhanbad

Reconstructed Question Structure & Topic Pattern

Section	Questions	Format
Aptitude / Probability / Statistics / Quant / Electrical Core	20	MCQs
SQL	2	SQL coding / query writing
Python / Pandas	3 + 2	3 MCQs + 2 coding questions
Total	27	All sections combined

Time limit: 90 minutes for the complete OA.

Section 1 — Aptitude / Probability / Statistics / Quant / Electrical Core

20 MCQs

Questions remembered from the OA

Q1 — Binomial Distribution:
If the probability that an archer will strike the target is **30%**, and the archer makes **10 attempts**, what are the **mean and variance** of the number of successful attempts?

Options shown included: **3, 2.1**; **3, 2.4**; **7, 2.1**; **7, 2.4**.

Q2 — Factorial / Number Theory:
What is the **highest power of 24** that divides **150!** exactly?

Options shown included: **42, 44, 46, 48**.

Q3 — Normal Distribution:
Given a dataset distributed normally with a **mean (μ) of 50** and a **standard deviation (σ) of 5**, what is the probability of selecting a sample with a **mean greater than 55**?

Options shown included: **Approximately 2.28%**, **15.87%**, **50%**, and **97.72%**.

Q4 — Cube / Painted Faces:
A cube of side **18 cm** has **two opposite faces coloured red** and the **other four faces coloured blue**. The cube is then broken into **729 equal-sized smaller cubes**. How many of the smaller cubes have **at least one red face and at least one blue face**?

Q5 — Probability / Bayes-type Question:
A girl knows that the probability that it **snows on any given night is 1/8**. Her father tells her that it **snowed last night**. Her second daughter says that the **father lies with probability 5/6**. Based on this information, what is the probability that it actually **snowed last night**?

Pattern of the other questions in the 20-MCQ section

- **Probability:** basic probability, conditional probability, Bayes/probability-of-events and binomial-type questions.
- **Statistics:** mean, variance, standard deviation, normal distribution/z-scores, hypothesis testing, null and alternative hypotheses, Type I/Type II errors and p-values.
- **Quantitative aptitude:** upstream/downstream and boat-stream questions, time-speed-distance, percentages, ratios/averages, divisibility and factorial-based questions.
- **Logical/visual reasoning:** cube-cutting, painted/coloured-face counting and spatial reasoning.

- **Electrical core:** basic electrical and circuit concepts, current/voltage/resistance and simple conceptual or numerical questions.

Section 2 — SQL

2 SQL coding/query questions

SQL Q1 — Email Campaign Analysis:
You are given two tables. The first table contains campaign information with **id** and **campaign_name**. The second table contains email activity with **campaign_id**, **emails_sent**, and **emails_opened**. Write an SQL query to output, for each campaign: **campaign_name**, **total_emails_sent**, **total_emails_opened**, and **total_emails_not_opened**. The result should contain one row per campaign.

Sample Input — campaigns

id	campaign_name
1	Summer Sale
2	Diwali Offer
3	New User Promo

Sample Input — email_campaigns

campaign_id	emails_sent	emails_opened
1	1000	700
1	500	300
2	2000	1200
2	1000	600
3	800	500

Expected Output

campaign_name	total_emails_sent	total_emails_opened	total_emails_not_opened
Summer Sale	1500	1000	500
Diwali Offer	3000	1800	1200
New User Promo	800	500	300

SQL Q2 — Device Traffic and Cost:
You are given two tables. The first contains device information with **id**, **mac** (MAC address), and **tariff**. The second contains traffic records with **id**, **dt** (date of traffic), and **amount**. Join the tables using the device ID and write an SQL query to output, for each MAC address, the **total traffic during May** as **traffic** and the corresponding **cost**, calculated using the applicable tariff and traffic amount. **Round the cost to 2 decimal places.** The output should contain one row per MAC address.

Sample Input — devices

id	mac	tariff
1	AA:BB:CC:DD:EE:01	0.50
2	AA:BB:CC:DD:EE:02	0.60
3	AA:BB:CC:DD:EE:03	0.40

Sample Input — traffic

id	dt	amount
1	2026-05-03	100
1	2026-05-15	50
1	2026-06-02	200
2	2026-05-10	200
2	2026-05-25	120
3	2026-05-18	150

Expected Output

mac	traffic	cost
AA:BB:CC:DD:EE:01	150	75.00
AA:BB:CC:DD:EE:02	320	192.00
AA:BB:CC:DD:EE:03	150	60.00

Section 3 — Python / Pandas

3 MCQs + 2 coding questions

- **MCQs:** mainly short Python/Pandas code-output or concept questions.
- **Pandas topics:** DataFrame creation/selection, loc/iloc, Boolean filtering, column operations, missing values, groupby/aggregation, transform/filter, data-type conversion, dates, categorical data and functions such as pd.cut().
- **Coding:** practical tabular-data manipulation such as loading data, cleaning/filtering, creating or transforming columns, grouping/aggregation and basic feature engineering.
- **Difficulty:** the 2 coding questions were remembered as easier than the longer customer-segmentation/data-analysis task shown in the reference material.

Accuracy note: The five questions in Section 1 are the questions remembered specifically from the OA. The remaining questions are represented by their topic/pattern rather than invented as exact wording. The SQL questions are written out in full from the remembered table structures and required outputs.

Overall structure: 20 aptitude/statistics/quant/electrical-core MCQs + 2 SQL questions + 3 Python/Pandas MCQs + 2 Python/Pandas coding questions, all within 90 minutes.